Class 6:

Evaluate the following integrals:

1. $\int_{-c}^{c}\int_{-b}^{b}\int_{-a}^{a}\left(x^2+y^2+z^2\right)dxdydz$ ans: $\frac{8abc}{3}\left(a^2+b^2+c^2\right)$
2. $\int_{0}^{a}\int_{0}^{x}\int_{0}^{x+y} e^{x+y+z}dzdydx$ ans: $\frac{1}{8}e^{4a}-\frac{3}{4}e^{2a}+e^{a}-\frac{3}{8}$
3. $\int_{0}^{\frac{\pi}{2}}\int_{0}^{a\sin\theta}\int_{0}^{\frac{a^2-r^2}{a}} rdzdrd\theta$ ans: $\frac{5\pi a^3}{64}$

****